SoleSense

Hardware Materials & How They're Used in the Product — Innovators Competition 2026, Stage 1 Implementation Plan (Team RANSAM)

SoleSense is a handheld, calibrated-force sensation screening device for diabetic peripheral neuropathy, expandable to vibration testing. This document consolidates every hardware material and component named in the Stage 1 Implementation Plan, organized by function, and explains how each one is used in the product.

1. Core Sensing Hardware

The sensing layer is the physical probe and its force-selection mechanism, plus the foot-positioning template — together these deliver the same calibrated force to the same anatomical point every time.

Material / Component	Spec	How it's used
Calibrated force elements (3x)	1 gf / 10 gf / 75 gf, matched to monofilament gauges 4.17 / 5.07 / 6.10	Mounted in the rotating carrier; each is indexed onto the probe axis in turn to deliver a graded, clinically referenced stimulus to the foot.
Reference digital force gauge (or calibrated scale)	Bench calibration instrument	Used in the pre-use calibration check to verify each force element's real output against its nominal target before testing begins, and periodically thereafter to track drift.
Force-selection servo	SG90 micro servo, ~1.8 kgf-cm stall torque	Rotates the 3-position carrier 120° between indexed positions to bring the selected force element onto the probe axis (torque margin ~30x the estimated demand).
Probe guide	Linear bushing / axial guide	Constrains the probe's motion to a single controlled, repeatable contact axis so the stimulus is applied consistently.
Contact/travel sensor	SPDT microswitch (lever type)	Confirms the probe has reached full travel/contact before a response is accepted; if contact is not detected, firmware requests a retry instead of logging a false 'not felt' result.

2. Electronics & Control Hardware

The control layer runs the fixed test sequence, reads patient input, gives on-device feedback, and logs results; the power system keeps it portable and cordless.

Material / Component	Spec	How it's used
Microcontroller	ESP32-WROOM-32 DevKit (38-pin)	Central controller: runs the test-sequence state machine, drives the servo, reads the contact sensor and response button, updates the OLED, times reactions, and writes results to the microSD card.
Display	SSD1306 0.96" I2C OLED, 128x64	Shows live test status — current test point and force level (e.g. "Point 3, Level 10gf") — over I2C (SDA-GPIO21, SCL-GPIO22).

Material / Component	Spec	How it's used
Response button	6 mm tactile push-button + cap	Patient presses this the instant a stimulus is felt; wired to GPIO4 with a 10 kOhm pull-down resistor for a clean logic-low when unpressed. Press timing yields the reaction-time measurement.
Storage interface	MicroSD SPI breakout module	Bridges the ESP32 to the microSD card over SPI (MOSI-GPIO23, MISO-GPIO19, SCK-GPIO18, CS-GPIO5).
Storage media	MicroSD card, 8 GB, Class 10	Stores a structured CSV log — one row per test event (timestamp, point ID, force level, response, reaction time) — so a session is fully reconstructable and feeds the phone/web dashboard.
Battery cell	18650 Li-ion cell, 2200-3000 mAh	Primary power source for cordless, portable operation; supports an estimated 15-20 hours of active testing per charge.
Charge / protection module	TP4056 USB-C Li-ion charger w/ protection	Handles USB-C charging and over-charge/over-discharge protection for the battery cell, kept as a separate stage from voltage regulation.
Voltage regulator	AMS1117-3.3 (or onboard DevKit regulator)	Produces the clean 3.3V/5V power rail that supplies the ESP32, OLED, microSD module, and servo independent of battery voltage sag.
Pull-down resistor	10 kOhm, 1/4 W	Sets a defined logic-low on the response-button GPIO input when the button is not pressed.

3. 3D-Printed Components

All printed parts are produced by FDM at FabLab UAE. PETG is used for parts handled repeatedly and cleaned between uses (better durability/chemical resistance); PLA is used for lower-stress trim where print speed matters more.

Component	Material / Print Settings	Est. Time	How it's used
Handle shell (top + bottom)	PETG, 0.20 mm layer, 20% gyroid infill	~4-5 hrs combined	Ergonomic housing that holds all electronics; snap-fit tabs with ~0.15 mm clearance for repeatable assembly/disassembly.
Force-element carrier	PETG, 0.12 mm layer, 40% (near-solid) infill	~1.5 hrs	Rotates between the 1/10/75 gf calibrated elements; detent bumps sized with ~0.10 mm interference give a positive tactile click at each indexed position.
Swappable tip socket	PETG, 0.16 mm layer, 60% infill	~30 min	Press-fit interface (bore undersized ~0.20 mm vs. the tip shaft) that lets the optional stretch-goal vibration tip share the same handle and template.
Button cap / OLED bezel	PLA, 0.20 mm layer, 15% infill	~20 min	Snap-fit UI trim (~0.20 mm clearance) around the response button and OLED; PLA chosen for stiffness and dimensional accuracy on small features.

4. Laser-Cut Components

Material / Component	Spec	How it's used
Foot positioning template	3 mm acrylic or plywood, laser cut, ~10-15 min	Fixes the 7 standardized anatomical test points (great toe, 1st/3rd/5th metatarsal, lateral/medial midfoot, heel) so the same point is tested in the same place every session, directly closing the 'no fixed positional guide' gap in the manual exam.

5. Assembly & Wiring Hardware

Material / Component	Spec	How it's used
Wiring & hardware	JST connectors, M2 screws, hook-up wire	General interconnects and fasteners used to assemble the housing shells, PCB, servo, and battery into one handheld unit.

6. Full Bill of Materials (Per Unit)

Consolidated from Table 9 (detailed BOM) and Table 4 (materials/equipment) of the Implementation Plan. Costs exclude the three calibrated force elements, which are sourced and verified separately as part of the calibration process.

Component	Model / Part	Qty	Unit Cost
Microcontroller	ESP32-WROOM-32 DevKit (38-pin)	1	~$8.00
Force-selection servo	SG90 micro servo (~1.8 kgf-cm)	1	~$3.00
Display	SSD1306 0.96" I2C OLED, 128x64	1	~$4.00
Storage interface	MicroSD SPI breakout module	1	~$2.00
Storage media	MicroSD card, 8 GB, Class 10	1	~$5.00
Battery cell	18650 Li-ion cell, 2200-3000 mAh	1	~$5.00
Charge / protection module	TP4056 USB-C Li-ion charger w/ protection	1	~$1.50
Voltage regulator	AMS1117-3.3 (or onboard DevKit regulator)	1	~$0.50
Response button	6 mm tactile push-button + cap	1	~$0.30
Contact/travel sensor	SPDT microswitch (lever type)	1	~$0.50
Pull-down resistor	10 kOhm, 1/4 W	1	~$0.05
Housing shells	3D-printed PETG (FDM), top + bottom	1 set	~$3.00
Force-element carrier	3D-printed PETG (FDM), fine layer height	1	~$1.00
Foot positioning template	Laser-cut 3 mm acrylic	1	~$4.00
Wiring & hardware	JST connectors, M2 screws, hook-up wire	1 set	~$3.00

Total estimated electronics + fabrication cost: ~$40.50 per unit, excluding the calibrated force elements (1 gf / 10 gf / 75 gf), which are sourced and verified against a reference force gauge as part of the calibration process (Section 1.5 of the plan).

Stretch-Goal Materials (Optional Vibration Expansion)

Material / Component	Spec	How it's used
Small vibration motor	Coin/ERM type	Optional second probe tip for a VPT-style vibration test, added only if the core calibrated-force build stays on schedule.

Source: SoleSense Stage 1 Implementation Plan (Sections 2.2, 4.3, 4.8, 4.17, 4.19) — Team RANSAM, submitted to FabLab UAE / Hamdan Bin Rashid Al Maktoum Foundation.